Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2023
Course Code: EEE 475, Session: 2019-2020
Course Title: RF and Microwave Engineering

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Group A

*[Answer any **two** of the following questions]*

1. (a) Write short notes on the following topics
   i. Directivity
   ii. Effective Height
   iii. Beam Area 9

   (b) What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna. 6

   (c) Describe how passive remote sensing is used to detect object. 5

   (d) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$. 5

2. (a) Show that the normalized resistances represent circles in reflection coefficient plane. 6

   (b) Write the properties of r-circles and x-circles of a Smith chart. 6

   (c) Describe how TE mode is different from TM mode in a circular waveguide. 3

   (d) The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15 + j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit [*sic*: should be short-circuit] is replaced by a load impedance $Z_L = 37 + j45$ Ω. 10

3. (a) Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example. 6

   (b) Explain the condition for constant gain for amplifier. 5

   (c) For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation. 5

   (d) A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S. 9

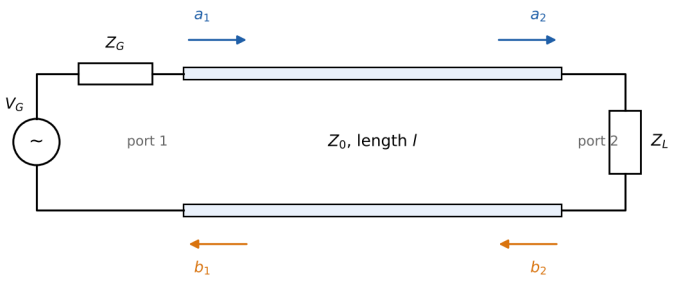


Figure 1 of the paper (redrawn)

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Group B

*[Answer any **two** of the following questions]*

4. (a) Write short notes on the following topic

   i. Reflection coefficient
   ii. Standing wave ratio
   iii. RF radiation hazards and safety standard
   iv. Microwave cavity resonators 8

   (b) Show how quarter wavelength line is used for matching. 4

   (c) Explain how the RADAR range equation is used to detect object. 7

   (d) A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line. 6

5. (a) Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave. 6

   (b) Find the forward voltage gain of a high-pass filter. 6

   (c) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. 7

   (d) Define near field and far field for an antenna. 7

6. (a) Explain how radio frequency affects the inductor and capacitor reactance. 6

   (b) From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient. 9

   (c) Briefly describe input impedance for open circuit termination and short circuit termination. 4

   (d) The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L. 7

Some useful Equations

$G_s = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$
$g_s = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}\,(1 - \lvert S_{11}\rvert^2)$
$d_{gi} = \dfrac{g_i S_{11}^*}{1 - \lvert S_{ii}\rvert^2(1 - g_i)}$
$r_{gi} = \dfrac{\sqrt{1 - g_i}\,(1 - \lvert S_{ii}\rvert^2)}{1 - \lvert S_{ii}\rvert^2(1 - g_i)}$

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